STAT 618
Bayesian Statistics

Quiz # 3

February 14, 2018

Name: SOLUTION

Center for Disease Control needs to estimate θ, the proportion of patients suffering from flu who were vaccinated earlier during the season. A sample of patients $\boldsymbol{X} = (X_1, \ldots, X_n)$ is collected, where

$$X_i = \begin{cases} 1 & \text{if} \quad \text{patient } i \text{ was vaccinated} \\ 0 & \text{if} \quad \text{patient } i \text{ was not vaccinated} \end{cases}$$

Without a prior information about θ, a non-informative improper prior is used,

$$\pi(\theta) = \frac{1}{\theta(1-\theta)}, \qquad \text{where } 0 < \theta < 1.$$

(a) Find the posterior mean and the posterior variance of θ.
(b) Describe all samples $\boldsymbol{X}$ that do not lead to a prper posterior distribution under the given prior.
An inventory of distributions is given on the back.

SOLUTION.

Bernoulli model... $f(\boldsymbol{X} \mid \theta) = f(X_1, \ldots, X_n \mid \theta) = \prod_{i=1}^{n} \theta^{X_i}(1-\theta)^{1-X_i} = \theta^{\sum X_i}(1-\theta)^{n-\sum X_i}$

Non-informative prior... $\pi(\theta) \sim \dfrac{1}{\theta(1-\theta)}$

Posterior... $\pi(\theta \mid \boldsymbol{X}) \sim f(\boldsymbol{X} \mid \theta)\pi(\theta) \sim \dfrac{1}{\theta(1-\theta)} \cdot \theta^{\sum X_i}(1-\theta)^{n-\sum X_i} = \theta^{\sum X_i - 1}(1-\theta)^{n-\sum X_i - 1}$,
which is a Beta(α_x, β_x) distribution with parameters $\alpha_x = \sum X_i$ and $\beta_x = n - \sum X_i$, if both $\alpha_x > 0$ and $\beta_x > 0$.

From the characteristics of Beta distribution, the posterior mean and variance are

$$\begin{aligned} \mathbf{E}\{\theta \mid \boldsymbol{X}\} &= \frac{\alpha_x}{\alpha_x + \beta_x} = \frac{\sum X_i}{n} = \overline{X} \\ \mathrm{Var}\{\theta \mid \boldsymbol{X}\} &= \frac{\alpha_x \beta_x}{(\alpha_x + \beta_x)^2(\alpha_x + \beta_x + 1)} = \frac{\sum X_i (n - \sum X_i)}{n^2(n+1)} = \frac{\overline{X}(1-\overline{X})}{n+1} \end{aligned}$$

(b) There is no proper posterior distribution when $\alpha_x = \sum X_i = 0$ or $\beta_x = n - \sum X_i = 0$, because both parameters of Beta distribution must be positive. This happens when either none of the sampled patients had a flu shot, and all $X_i \equiv 0$, or all the sampled patients had a flu shot, in which case all $X_i \equiv 1$.

This does make sense! The problem states that θ is strictly between 0 and 1, i.e., there are patients in the flu infected population who had a flu shot and those who did not have it. Therefore, there is not enough information from the extreme data $X_i \equiv 0$ and $X_i \equiv 1$ to come up with a reasonable estimator of θ between 0 and 1. With no information from the prior as well, we ened up with no proper posterior, and no decision.

However, for any other data, with $0 < \sum X_i < 1$, the posterior mean coincides with the sample mean $\mathbf{E}\{\theta \mid \boldsymbol{X}\}\overline{X}$. That is, in the absence of an informative prior, the Bayes rule is the standard frequentist estimator.

Inventory of some distributions

Discrete distributions

Distribution	Parameters	Probability mass function	$\mathbf{E}(X)$	$\mathrm{Var}(X)$	Comments
Binomial $\mathcal{B}(n,p)$	$n=1,2,\ldots$ $p\in(0,1)$	$\binom{n}{x}p^x(1-p)^{n-x}$, $x=0,1,\ldots,n$	np	$np(1-p)$	
Geometric $\mathcal{G}e(p)$	$p\in(0,1)$	$(1-p)^{x-1}p,\ x=1,2,\ldots$	$1/p$	$(1-p)/p^2$	$\mathcal{NB}(1,p)$
Negative Binomial $\mathcal{NB}(k,p)$	$k=1,2,\ldots$ $p\in(0,1)$	$\binom{x-1}{r-1}p^k(1-p)^{x-k}$, $x=k,k+1,\ldots$	$1/p$	$(1-p)/p^2$	
Poisson $\mathcal{P}(\lambda)$	$\lambda>0$	$e^{-\lambda}\lambda^x/x!,\ x=0,1,2,\ldots$	λ	λ	

Continuous distributions

Distribution	Parameters	Density	$\mathbf{E}(X)$	$\mathrm{Var}(X)$	Comments
Normal $\mathcal{N}(\mu,\sigma)$	$\mu\in R$ $\sigma>0$	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}},\ x\in R$	μ	σ^2	
Uniform $\mathcal{U}(a,b)$	$a<b$	$\frac{1}{b-a},\ a<x<b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$Be(1,1)$
Gamma $\mathcal{G}(\alpha,\beta)$	$\alpha,\beta>0$	$\frac{1}{\Gamma(\alpha)\beta^\alpha}x^{\alpha-1}e^{-x/\beta}$ $x>0$	$\alpha\beta$	$\alpha\beta^2$	
Exponential $\mathcal{E}(\beta)$	$\beta>0$	$\frac{1}{\beta}e^{-x/\beta},\ x>0$	β	β^2	$\mathcal{G}(1,\beta)$
Chi-square $\chi^2(\nu)$	$\nu>0$ (d.f.)	$\frac{1}{2^{\nu/2}\Gamma(n/2)}x^{\nu/2-1}e^{-x/2}$ $x>0$	ν	2ν	$\mathcal{G}(\nu/2,2)$
Beta $\mathcal{B}e(\alpha,\beta)$	$\alpha,\beta>0$	$\frac{\Gamma(\alpha,\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$ $0<x<1$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	
Cauchy $\mathcal{C}(\alpha,\beta)$	$\alpha\in R$ $\beta>0$	$\frac{\beta}{\pi[\beta^2+(x-\alpha)^2]},x\in R$	$-$	$-$	
Inverse Gamma $\mathcal{IG}(\alpha,\beta)$	$\alpha,\beta>0$	$\frac{1}{\Gamma(\alpha)\beta^\alpha x^{\alpha+1}}e^{-1/x\beta}$ $x>0$	$\frac{1}{\beta(\alpha-1)}$ if $\alpha>1$	$\frac{1}{\beta^2(\alpha-1)^2(\alpha-2)}$ if $\alpha>2$	
Pareto $\mathcal{P}a(x_0,\alpha)$	$\alpha,x_0>0$	$\frac{\alpha x_0^\alpha}{x^{\alpha+1}},\ x>x_0$	$\frac{\alpha x_0}{\alpha-1}$ if $\alpha>1$	$\frac{\alpha x_0^2}{(\alpha-1)^2(\alpha-2)}$ if $\alpha>2$	